Principles, Practices, and Patterns, by Vladimir Khorikov (Manning, 2020). *Effective Software Testing: A Developers Guide*, by Maurício Aniche (Manning, 2022), uses Java examples but covers a broad range of topics and techniques. Alternatively, for an in-depth look at testing with xUnit in .NET Core, see *.NET in Action*, 2nd ed., by Dustin Metzgar (Manning, 2023).

In section 35.1 I introduce the .NET software development kit (SDK) testing framework and show how you can use it to create unit testing apps. I describe the components involved, including the testing SDK and the testing frameworks themselves, like xUnit and MSTest. Finally, I cover some of the terminology I use throughout this chapter and chapter 36.

This chapter focuses on the mechanics of getting started with xUnit. You'll learn how to create unit test projects, reference classes in other projects, and run tests with Visual Studio or the .NET command-line interface (CLI). You'll create a test project and use it to test the behavior of a basic currency-converter service. Finally, you'll write some simple unit tests that check whether the service returns the expected results and throws exceptions when you expect it to.

Let's start by looking at the overall testing landscape for ASP.NET Core, the options available to you, and the components involved.

35.1 An introduction to testing in ASP.NET Core

In this section you'll learn about the basics of testing in ASP.NET Core. You'll learn about the types of tests you can write, such as unit tests and integration tests, and why you should write both types. Finally, you'll see how testing fits into ASP.NET Core.